CELESTIAL MECHANICS × AI · 2026

# From the next token to the next theorem

How a model works — and how we should use it

Bowen Liu

Draft v0.1

45 min

# What the model sees: ten integers

EN · ONE SENTENCE

Celestial mechanics is the poetry of orbits.

| Cel | estial | _mechanics | _is | _the | _poetry | _of | _or | bits | . |
|---|---|---|---|---|---|---|---|---|---|
| 69126 | 90010 | 52347 | 382 | 290 | 38153 | 328 | 503 | 24043 | 13 |

**46** characters → **10** tokens → 10 integers · _ marks a leading space

CHINESE · FOUNDATIONS OF CELESTIAL MECHANICS

本书系统地介绍了天体力学

| 本 | 书 | 系统 | 地 | 介绍 | 了 | 天 | 体 | 力 | 学 |
|---|---|---|---|---|---|---|---|---|---|
| 4087 | 20121 | 31936 | 5557 | 37887 | 4531 | 867 | 7074 | 11343 | 8590 |

**12** characters → **10** tokens → 10 integers · two-character words often map to one token

“This book systematically introduces celestial mechanics.”

01 Where do these integers come from?

02 How does it predict the next one?

03 How do we put it to work?

# CoT: write the thinking down

Scope: verbal and logical reasoning; arithmetic is a separate step.

Question

A period of 2 years; the semi-major axis grows 4×. What is the new period?

DIRECT ANSWER

8 years ✗ — a number without a plan

COT · PLAN

$T \propto a^{3/2}$; $T = 2 \times 4^{3/2}$

plan ✓ · arithmetic next

TOOL · COMPUTE

```
run_python("2 * 4**1.5") → 16.0 ✓
```

WHAT COT DOES

Turns words into a plan: parse, recall, set up the relation

WHAT REMAINS

Arithmetic: a separate, tool-friendly step

ESSENCE

Explicit steps; exact numbers are a separate skill

> CoT writes the plan; the tool computes the number.

# Put the model back in the system




Inner ring: one call · Middle ring: across calls · Outer ring: engineering boundary

| | | |
|---|---|---|
| 01 | **LLM** | A probability generator: tokens in, a distribution over the next token out |
| 02 | **Prompt** | This call's conditions: system / user / few-shot |
| 03 | **Context** | Every token visible to this call; the window is finite |
| 04 | **Memory** | State beyond the window: write / retrieve / compress / forget |
| 05 | **Agent loop** | Observe → act → observe; state lives outside |
| 06 | **Harness** | Tools, sandbox, permissions, verification, logs |

> The model is the CPU; prompt, context, memory, loop and harness are the runtime.

# Acceleration: from basic use to gold medals

2023 · BEFORE

## Basic use

- Explain concepts, search the literature, write code
- Do simple calculations, often wrong
- Complex proofs still need people
- Role: assistant

→ 19 months

2025 · AFTER

## Research level

- IMO 2025 gold level 35/42
- Lean proofs, machine-checkable
- OpenAI 2026: 372 problems, ≈3 h each (report)
- Role: candidate generator

2024.07 **28/42** silver → 2025.07 **35/42** gold level — 12 months, +7 points

> This is a staircase with **three steps**.

# Timeline: three steps

2023–24

## Tools: search + verification

2023.12
**FunSearch**
LLM + evaluator + evolutionary search; a new cap-set construction

2024.01
**AlphaGeometry**
Neuro-symbolic; IMO geometry 25/30

2024.07
**AlphaProof**
Lean + RL; IMO 2024 silver 28/42 (+AG2)

2024–25

## Reasoning: RL + test-time compute

2024.09
**OpenAI o1**
RL teaches thinking; test-time compute

2025.01
**DeepSeek-R1**
RLVR; reasoning behavior emerges; open weights

2025.05
**AlphaEvolve**
Evolutionary coding agent; 4×4 complex matrix product in 48 multiplications

2025–26

## Scale: from benchmarks to open problems

2025.07
**IMO 2025**
Gold level 35/42; natural language, no tools

2025
**Formal gold (claimed)**
Aristotle et al.; Lean-checkable

2026.05–10
**OpenAI**
Unit-distance counterexample; NS / Riemann-related (claimed); 372 problems (report)

> Every leap combines a **generator, a verifier and search**.

# The motor and the assembly line

A story Eric Schmidt tells — and the AI parallel.

ELECTRIFICATION · SAME LAYOUT **+30%**



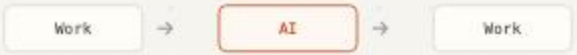

The motor replaces the steam engine; everything else stays.

ELECTRIFICATION · NEW LAYOUT **10×+**



Small motors per machine; the factory is rebuilt around flow.

AI · SAME WORKFLOW **+30%**




Ask AI inside the old step: polish, slides, autocomplete.

AI · NEW WORKFLOW **10×+**




Agents, tools, context, verification, loop.

> The gain is in the **layout**.

Most teams stop at +30%: AI sits inside the old step.

# The algorithm engineer's dilemma

It arrived about a year ago. Mathematics is next.

"If the model can write the code, what is my job?"

BEFORE AGENTS

## Write the algorithm

- Implement, debug, optimize
- Own the code
- Value = code shipped

→ ~1 year

WITH AGENTS

## Direct the search

- Frame the problem
- Design the harness
- Verify and own the system

> The job moves from producing answers to **directing the search**.

# From the chat box to the working directory

The model stays the same. The harness moves closer to the work.

2022–23

### Web chat

**Ask and copy**
One question, one answer

**Text only**
The model describes; you execute

2023–24

### IDE copilot

**Complete in the editor**
Suggestions; the human runs the code

**Context = open files**
A faster editor, same loop

2024–26

### CLI / desktop agent

**Read, write, run**
Files, shell, tests, git

**Observe → act → observe**
Permissions and sandbox

- OpenCode
- Claude Code
- Codex CLI
- Cursor
- ChatGPT / Claude desktop

> Same model. Different **harness**.

In a research repo, the agent can run the experiment and return the numbers.

# From the next token to the next theorem

`token` → `probability` → `candidate` → `verification` → **`theorem`**

Probability gives breadth

Verification gives correctness

**People set the direction**

> The next theorem still begins with the **next token**.